Inferring Support from Endorsement Experiments*

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Abstract

Endorsement experiments measure how group endorsements affect policy support, and researchers routinely interpret the sign of the average treatment effect as indicating "sentiment" toward the endorsing group. I show that this interpretation does not follow when applied to levels of support: the ATE reflects the balance of switchers, not the proportion of supporters. Using both the treatment effect and the baseline support level, I derive bounds on the proportion of supporters under a family of behavioral models. Applied to Blair et al. (2013), where baseline policy support is 74% and the treatment effect is −1.1 percentage points, the estimates range from 39% to 99% depending on assumptions about individual-level switching behavior, with the binary switching model yielding 72% and a discrete-scale model that accounts for ceiling effects on the 5-point response scale yielding 59%. No model considered here supports their characterization of "low regard" for militant groups. The precise level of support depends on assumptions the data cannot adjudicate, but the broader point is that endorsement experiments are underidentified for the quantity researchers most want to estimate—the proportion of supporters—without strong auxiliary assumptions about individual behavior.

1 Introduction

Blair et al. (2013) conclude that Pakistanis hold militant groups in "low regard" based on small negative treatment effects from endorsement experiments. This inference—from the sign of the ATE to the level of support—does not follow without additional assumptions about individual switching behavior. The sign of the average treatment effect (ATE) tells us the balance of switchers, not the level of support for the endorsing group. With high baseline policy support, a small negative effect is consistent with a wide range of supporter proportions, including large majorities.

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The problem is general. Endorsement experiments identify the ATE under standard exclusion restrictions, but the ATE alone does not identify the proportion of supporters. Mapping from the ATE to a supporter proportion requires auxiliary assumptions about individual switching behavior. Under binary switching (every supporter moves to 1, every opponent moves to 0), the treatment group average equals the supporter proportion exactly. Under symmetric small effects (± 0.05), the same ATE implies only 39% support. The data cannot distinguish these models.

This paper develops a framework for what endorsement experiments can and cannot tell us about support levels. The framework uses both the treatment effect and the baseline support level (the regression constant), and yields bounds under different assumptions about individual behavior. The key insight is simple: baseline support information is crucial and routinely ignored, and the range of estimates compatible with a given ATE is wide enough that "sentiment" interpretations based on the sign alone are unreliable.

Applied to Blair et al. (2013), where baseline support is 74% and the treatment effect is -1.1 percentage points, the framework yields estimates ranging from 39% (symmetric effects) to 99% (extreme opponent reaction), with the binary switching model at 72%. None of these estimates support their characterization of "low regard." But the 72% figure requires a strong behavioral assumption that is testable and likely violated in their data.

This paper proceeds as follows. Section 2 reviews endorsement experiments and the interpretation problem. Section 3 develops the inferential framework. Section 4 reanalyzes Blair et al.. Section 5 discusses implications for practitioners. Section 6 concludes.

2 Background: Endorsement Experiments

2.1 Design

Endorsement experiments measure latent attitudes toward controversial actors by randomizing whether a group endorses a policy (Bullock et al., 2011). In a standard between-subjects design, respondents see either:

- **Treatment**: "Do you support this policy? [Group Y endorses this policy.]"
- **Control**: "Do you support this policy?"

The identifying assumption is that the endorsement affects policy support only through respondents' attitudes toward the endorsing group. Supporters of the group should increase their policy support; opponents should decrease theirs. The design has been applied to measure support for militant groups in Pakistan (Blair et al., 2013), combatant support in Afghanistan (Lyall et al., 2013), and attitudes in other conflict settings (Blair et al., 2014).

2.2 Standard Interpretation and Its Limits

Researchers typically report the average treatment effect and interpret its sign as indicating "sentiment" toward the endorsing group. A negative ATE is taken to mean respondents

view the group unfavorably; a positive ATE suggests favorable attitudes. Blair et al. (2013) exemplify this practice when they conclude that negative coefficients indicate Pakistanis hold militant groups "in low regard."

This interpretation conflates two distinct quantities: the net direction of attitude-driven switching and the proportion of the population holding each attitude. The ATE reflects the net effect of two countervailing forces: supporters increasing their policy support and opponents decreasing theirs. The sign tells us which force dominates on average, but not by how much, and not what proportion of the population falls into each category.

Consider an ATE of -0.01 (a 1 percentage point decrease in policy support). This could arise from 51% opponents each reducing support slightly and 49% supporters each increasing support slightly, or from 1% opponents dramatically reducing support with 99% unaffected, or from many other combinations. Without additional structure, the ATE alone cannot distinguish these scenarios.

2.3 The Identification Problem

The core difficulty is that endorsement experiments are designed to identify the ATE, not the proportion of supporters. Moving from the ATE to a supporter proportion requires a model of individual switching behavior, and different models yield very different answers. This is not a failure of the experimental design; it is a fundamental feature of the inferential problem. The ATE is a well-identified causal quantity. The supporter proportion is a structural parameter that requires additional assumptions.

3 A Framework for Inference

3.1 Setup

Consider respondents falling into two categories based on their attitudes toward the endorsing group: supporters (proportion p) who view the group favorably, and opponents (proportion $1-p$) who view it unfavorably. Let δ^+ denote the average treatment effect among supporters and δ^- the average effect among opponents, where we expect $\delta^+ > 0$ and $\delta^- < 0$.

The observed ATE decomposes as:

$$\beta = p \cdot \delta^+ + (1 - p) \cdot \delta^- \tag{1}$$

This equation has three unknowns (p, δ^+, δ^-) and one observed quantity (β). The system is underidentified—a special case of the partial identification problem studied by Manski (1990). Adding the baseline support level (the constant term α from the regression $\text{Support} = \alpha + \beta \cdot \text{Treatment} + \varepsilon$) provides a second equation, but the system remains underidentified without restrictions on δ^+ and δ^-.

3.2 Binary Switching

The strongest restriction sets δ^+ and δ^- such that all supporters end up at 1 and all opponents end up at 0 in the treatment group. Under this assumption, the treatment group average is:

$$\text{Treatment average} = p \cdot 1 + (1 - p) \cdot 0 = p \tag{2}$$

Since the treatment average equals $\alpha + \beta$, we get:

$$p = \alpha + \beta \tag{3}$$

The treatment group average equals the proportion of supporters. This result is exact under the binary switching model: the system has three unknowns, one equation from the ATE, and two constraints from binary switching (δ^+ and δ^- are pinned down by the assumption that outcomes are 0 or 1), leaving the system exactly identified.

3.3 Testable Implications of Binary Switching

Binary switching has a strong testable implication. If all treatment group responses are either 0 or 1, the treatment group variance must equal $p(1-p)$. With $p = 0.725$ (the Afghan Taliban treatment mean from the data), the predicted treatment variance is $0.725 \times 0.275 \approx 0.199$, and the standard deviation is approximately 0.45.

Blair et al. (2013) measured policy support on a 5-point scale rescaled to $[0, 1]$. If the treatment group standard deviation is substantially below 0.45, the binary switching model is rejected. The actual treatment standard deviation in the data is approximately 0.29, well below the predicted value and indicating that respondents are not clustering at the extremes of the scale.

This does not invalidate the framework. It means the precise mapping $p = \alpha + \beta$ does not hold, and the analyst must consider softer switching models.

3.4 Softer Switching Models

Without binary switching, the analyst faces the decomposition in Equation (1) with three unknowns. Different assumptions about δ^+ and δ^- yield different values of p.

Symmetric small effects. If $\delta^+ = +d$ and $\delta^- = -d$ for some common effect magnitude d:

$$p = \frac{\beta}{2d} + \frac{1}{2} \tag{4}$$

With $d = 0.05$ (a moderate shift on a $[0, 1]$ scale): $p = -0.011/0.1 + 0.5 = 0.39$, or 39%.

Asymmetric extreme reaction. If opponents react maximally ($\delta^- = -1$, moving to 0) while supporters are unaffected ($\delta^+ = 0$):

$$\beta = -(1 - p) \implies p = 1 + \beta \tag{5}$$

With $\beta = -0.011$: $p = 0.989$, or 99%.

Normal heterogeneity. If individual effects $\delta_i \sim \mathcal{N}(\beta, \sigma^2)$, the proportion with positive effects (loosely, "supporters" in a continuous sense) is:

$$P(\delta_i > 0) = 1 - \Phi\left(\frac{-\beta}{\sigma}\right) \tag{6}$$

With $\beta = -0.011$ and $\sigma = 0.05$: $P(\delta_i > 0) \approx 41\%$.

These models yield a wide range of implied support—from 37% to 99%—depending on what one assumes about individual switching behavior. Section 4 applies them to the Blair et al. data.

3.5 What Baseline Information Buys

The baseline (constant term) contributes in two ways. First, under binary switching, it provides the second equation needed for exact identification (Equation (3)). Without it, the binary framework yields only $p_{\max} = (\beta + 1)/2 = 49.4\%$, which is far from the 72.5% obtained with the constant.

Second, even without binary switching, the baseline constrains the plausible range. A control average of 74% means that 74% of respondents support the policy absent the endorsement cue. Any model of switching must be consistent with this fact. Models that imply very low baseline support are ruled out.

The contribution of baseline information is the most practically important finding of this paper. Many analyses of endorsement experiments report only the ATE and its sign. Reporting the treatment group average, or equivalently the constant and the ATE together, enables a much richer set of inferences. The estimate is also robust to partial awareness: if 30% of respondents have no opinion about the endorsing group and are unaffected by the endorsement, the implied support among opinion-holders under binary switching is $\beta/q + \alpha =$ 72%, essentially unchanged.

4 Reanalyzing Blair et al.

4.1 Data

Blair et al. (2013) conducted endorsement experiments in Pakistan measuring attitudes toward four militant groups (Afghan Taliban, Pakistani Taliban, Kashmiri militants, and sectarian organizations). Respondents were asked about support for four policies (polio vaccination, FCR reform, Durand Line, curriculum standardization) with and without militant group endorsements, on a 5-point scale rescaled to $[0, 1]$. The sample comprised approximately 5,200 respondents in a nationally representative survey. I reanalyze their original

data, available through the `endorse` R package (Bullock et al., 2011).[1]

Their interpretation has been influential. They write: "We find that Pakistanis in general are **weakly negative** toward Islamist militant organizations" and "The coefficients are negative and statistically significant at the 10% level for all but the sectarian tanzeems, suggesting that **Pakistanis hold militant groups in low regard**." This follows the standard practice of interpreting the ATE sign as a summary of group-level sentiment. The framework developed above shows why this step requires additional assumptions that, when made explicit, yield a wide range of support estimates.

4.2 Binary Switching and the Variance Test

Across all 16 policy × endorser combinations, the average control mean is $\alpha = 0.736$ and the average treatment effect is $\beta = -0.013$, giving an average treatment group mean of 0.723. Baseline support varies across policies: from 67% (Durand Line) to 83% (polio vaccination). Under binary switching, implied support ranges from 71.8% to 72.7% across groups.

But the binary switching model is testable, and the data reject it. Binary switching predicts a treatment group standard deviation of approximately 0.45; the actual treatment standard deviation is 0.29. The variance ratio—observed to predicted—is 0.42 across groups (Table 1). Respondents use the full range of the 5-point scale, not just the endpoints.

Table 1. Variance Test for Binary Switching

Group	Treat. Mean	Predicted SD	Actual SD	Var. Ratio
Afghan Taliban	0.725	0.447	0.287	0.413
Pakistani Taliban (TTP)	0.718	0.450	0.294	0.427
Kashmiri Militants	0.722	0.448	0.299	0.444
Sectarian Groups	0.727	0.445	0.291	0.428

Researchers can compute this diagnostic from standard regression output without microdata: given β, α, and the outcome standard deviation σ, the variance ratio is $\sigma^2/[(\alpha+\beta)(1-\alpha-\beta)]$. A ratio well below 1 rejects binary switching.

4.3 Ceiling Effects on the Discrete Scale

The 5-point scale creates a mechanical asymmetry that biases the ATE downward. With 42% of control responses at 1.0 and 29% at 0.75, supporters who are already at the ceiling cannot shift higher, while opponents at the ceiling can shift down freely. Over two-fifths of the positive-shift potential is lost to truncation.

[1] We use the unweighted microdata. Blair et al. use survey weights to account for oversampling in Balochistan and Khyber Pakhtunkhwa; our unweighted estimates differ slightly from their Table 1 but the qualitative conclusions are unchanged.

Formalizing: let each supporter shift up by s categories (capped at 1) and each opponent shift down by s categories (floored at 0). The effective average shifts, accounting for truncation, are $\bar{\delta}^+$ and $\bar{\delta}^-$. Solving $\beta = p\bar{\delta}^+ + (1-p)\bar{\delta}^-$ for p, a 1-category shift (± 0.25) yields 59% support despite the negative ATE (Table 2). The ceiling effect explains part of why the ATE is negative: supporters' positive reactions are mechanically attenuated by the scale.

Table 2. Ceiling Effects on the 5-Point Scale

Shift	Eff. δ^+	Eff. δ^-	Ceiling Loss	Implied p
1 category	0.145	-0.236	0.105	59.0%
2 category	0.218	-0.454	0.282	65.9%
3 category	0.251	-0.631	0.499	70.3%

4.4 Implied Support Across Models

Table 3 collects implied support under all models considered, from the softest (symmetric effects, 37%) to the strongest (extreme opponent reaction, 99%). Figure 1 and Figure 2 display the range graphically.

Table 3. Implied Support Under Different Behavioral Models

Model	Key Assumption	Implied Support
Binary switching (with baseline)	$\alpha + \beta$; $\alpha = 0.736$	72.5%
5-point scale, 2-step shift	Shift ± 2 categories, ceiling/floor	65.9%
5-point scale, 1-step shift	Shift ± 1 category, ceiling/floor	59.0%
Binary switching (no baseline)	$(\beta + 1)/2$, no constant	49.4%
Normal heterogeneity ($\sigma = 0.1$)	$\delta_i \sim \mathcal{N}(-0.011, 0.1^2)$	45.5%
Symmetric effects ($d = 0.05$)	$\delta^+ = +0.05$, $\delta^- = -0.05$	38.7%
Extreme opponent reaction	$\delta^- = -1$, $\delta^+ = 0$	98.9%

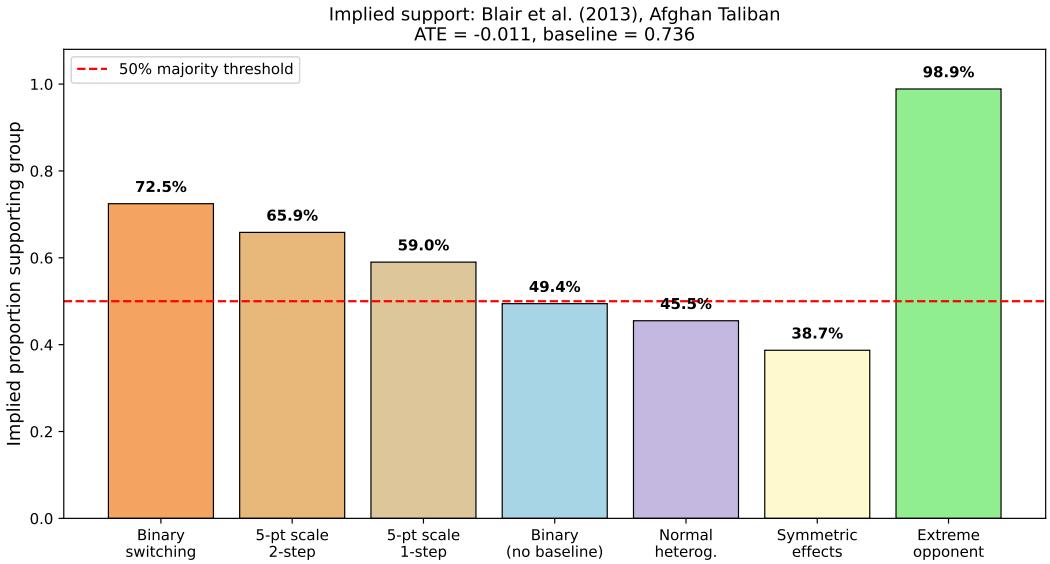

Figure 1. Implied support varies substantially across behavioral models. No single estimate is privileged without a defended choice of model.

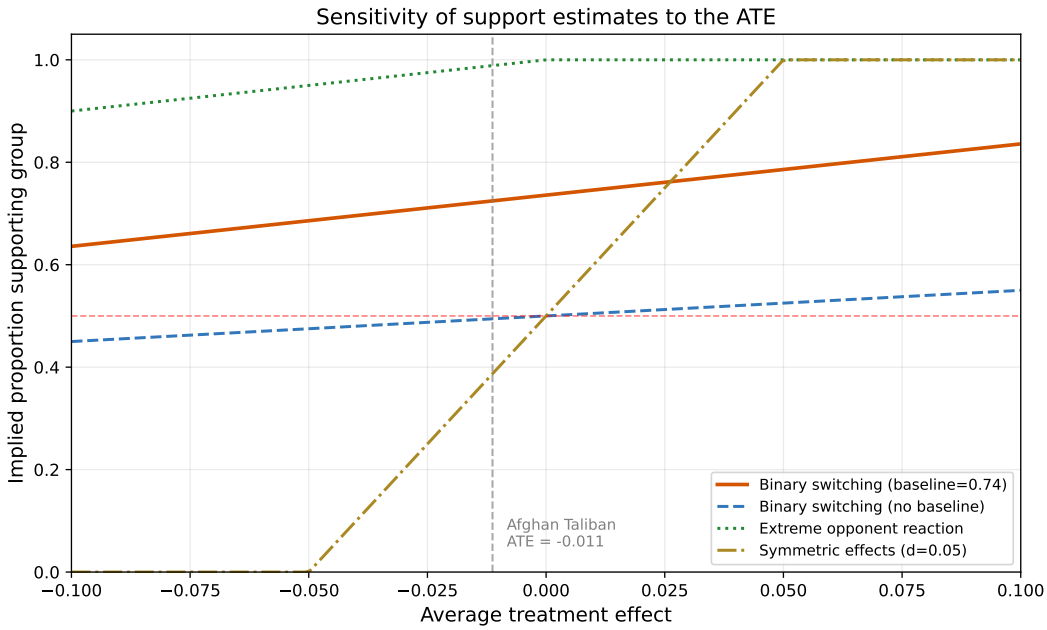

Figure 2. Sensitivity of support estimates to the ATE under different models.

The width of this range—37% to 99%—reflects the genuine underidentification of the problem. The data constrain the supporter proportion only loosely without a commitment to a specific behavioral model. But no model yields "low regard." Even the most conservative estimate (symmetric effects at 37%) describes a substantial minority, not negligible support. A model-free lower bound reinforces this: randomly pairing treatment and control

observations, the treatment response exceeds the control response 46% of the time. This requires no behavioral assumptions and falls in the middle of the parametric range.

4.5 External Validation

Direct polling on militant group support in Pakistan yields much lower figures. Pew Research Center surveys from this period find 8–13% explicit support for the Taliban and al Qaeda, with 33–40% declining to offer an opinion (Pew Research Center, 2013, 2014). The gap between 8–13% (direct polling) and the endorsement experiment estimates (37–72% depending on model) is large.

Part of this gap plausibly reflects social desirability bias and fear, which is precisely the motivation for indirect measurement. But the gap is wide enough to warrant caution. Even assigning every "don't know" respondent to the support camp yields at most roughly 60%, which is below the binary switching estimate but consistent with the discrete-scale model. The external evidence does not decisively adjudicate between models.

5 Discussion

Three recommendations follow from this analysis.

Report treatment group averages, not just ATEs. Under binary switching, the treatment average directly estimates the supporter proportion. Even when binary switching is implausible, the treatment average provides a useful anchor. Reporting only the ATE and its sign discards information.

Do not equate the ATE sign with "sentiment." The sign tells you which switching force dominates, not whether the group is popular or unpopular. With high baseline support, negative ATEs can coexist with majority support under multiple behavioral models.

Discuss identifying assumptions explicitly. Moving from ATEs to support levels requires a behavioral model. Authors should state which model they assume, defend it, and report how sensitive results are to alternatives. The variance test provides a simple diagnostic for the binary switching assumption.

6 Conclusion

Blair et al. (2013) conclude that Pakistanis hold militant groups in "low regard" based on small negative treatment effects. This inference does not follow. The sign of the ATE reflects the balance of switchers, not the level of support. Reanalyzing their original data, I derive a range of supporter proportions under different behavioral models: from 39% (symmetric effects) to 99% (extreme opponent reaction), with the binary switching model at 72%.

The characterization of "low regard" does not follow from any model considered here. But the precise level of support is sensitive to assumptions the data cannot adjudicate. The

binary switching estimate of 72% requires an assumption that the data reject on variance grounds. The symmetric effects estimate of 39%, while lower, still represents a substantial minority rather than negligible support. Two caveats apply: the interpretation of the constant as baseline support requires a correctly specified regression, and the framework assumes the endorsement manipulation is credible—if respondents doubt that militant groups endorse benign policies, the measured effects understate true attitudes.

The broader methodological point is that endorsement experiments identify average treatment effects well but are underidentified for the quantity researchers most want: the proportion of supporters. Moving from ATEs to support levels requires explicit behavioral assumptions. Researchers should report treatment group averages, discuss identifying assumptions, and present sensitivity analyses rather than drawing "sentiment" conclusions from the sign of the ATE alone.

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